

ORIGINAL CONTRIBUTION

Beyond the Threshold: Continuous Creatinine Change and Outcomes After Endovascular Thrombectomy for Acute Ischemic Stroke

Ghil Schwarz¹, MD; Angelo Cascio Rizzo², MD; Gareth Ambler, MD; Pawel Wrona³, MD; Agnieszka Slowik⁴, MD; Aleksandra Bodzioch, MD; Mohamed F. Doheim⁵, MD; Alhamza R. Al-Bayati⁶, MD; Raul G. Nogueira⁷, MD; Ana Paiva Nunes⁸, MD; Patricia Ferreira, MD; Matteo Paolucci⁹, MD, PhD; Andrea Zini¹⁰, MD; Luigi Simonetti¹¹, MD; Norbert Leško¹², MD, PhD; Jakub Fedorko¹³, MD; Zuzana Gdovinová¹⁴, MD, PhD; Luca Scarcia¹⁵, MD, MSc; Erwah Kalsoum¹⁶, MD; Firas Farhat, MD; Morin Beyeler¹⁷, MD; Adnan Mujanovic¹⁸, MD; Marcel Arnold¹⁹, MD; Torcato Meira²⁰, MD, PhD; Marta Morais, MD; Leonor Francisco²¹, MD; Marcin Wiącek²², MD, PhD; Paulina Pudło²³, MD; Halina Bartosik-Psujek, MD, PhD; Patricia Calleja²⁴, MD; Fernando Ostos²⁵, MD, PhD; David Seoane²⁶, MD; Negrita Anca²⁷, MD; Razvan Alexandru Radu²⁸, MD, PhD; Cristina Tiu²⁹, MD, PhD; Sami Al Kasab³⁰, MD; Ahmad Abu Qdais³¹, MD; Imad Samman Tahhan³², MD; Oscar Ayo-Martin³³, MD, PhD; Francisco Hernández-Fernández, MD, PhD; Maria Paya³⁴, MD; Juan David Molina-Nevo³⁵, MD; Beata Maria Labuz-Roszak³⁶, MD, PhD; Danilo Toni³⁷, MD; Karolina Moszko³⁸, MD; Mateusz Łukasz Roszak³⁹, MD; Manuela De Michele⁴⁰, MD; Elena Barrile⁴¹, MD; Prasanna Eswaradass⁴², MD, DM; Margaret L. Houghton⁴³, BSN, RN; Tiffany Barkley⁴⁴, DO; Amir Ali⁴⁵, MD; Jelle Demeestere⁴⁶, MD, PhD; Wayne Martin Bauknight⁴⁷, MD; Sunil Anil Sheth⁴⁸, MD; Louise Maes⁴⁹, MD; Anke Wouters⁵⁰, MD, PhD; João Pedro Marto⁵¹, MD, PhD; José Blazer Costa⁵², MD; Nicola Davide Loizzo⁵³, MD; Andrea Romi⁵⁴, MD; Carmelo Tiberio Currò⁵⁵, MD; Piotr Luchowski, MD, PhD; Maksymilian Seweryn⁵⁶, MD; Konrad Rejdak⁵⁷, MD, PhD; Piers Klein⁵⁸, MA; Mohamad Abdalkader⁵⁹, MD; Valentina Saia⁶⁰, MD, PhD; Antioco Sanna⁶¹, MD; Tiziana Tassinari⁶², MD; Maurizio Acampa⁶³, MD, PhD; Francesca Rosini, MD; Rossana Tassi⁶⁴, MD; Marta Ewa Bilik⁶⁵, MD, PhD; Anna Maria Bandzarewicz-Samcik⁶⁶, MD; Jiangyong Min⁶⁷, MD, PhD; Naser HajAissa⁶⁸, MD; Marilena Mangiardi⁶⁹, MD, PhD; Sabrina Anticoli⁷⁰, MD; Enrico Pampana⁷¹, MD; Carlos Hervás-Testal⁷², MD; Ricardo Rigual⁷³, MD, PhD; Blanca Fuentes⁷⁴, MD, PhD; Davide Strambo⁷⁵, MD, PhD; Carl Manuata Tetaria⁷⁶, MD; Guillaume Saliou⁷⁷, MD; Andrea Salvatore Caramma⁷⁸, MD; Davide Maimone⁷⁹, MD; Pier Andrea Rizzo⁸⁰, MD; Marco Moci⁸¹, MD; Irene Scala⁸², MD; Antonio Cruz-Culebras⁸³, MD; Rocío Vera Lechuga⁸⁴, MD, PhD; Sebastian Garcia Madrona⁸⁵, MD; Giovanni Merlino⁸⁶, MD, PhD; Mariarosaria Valente⁸⁷, MD; Arianna Todesco, MD; Manuel Bolognese⁸⁸, MD; Lehel-Barna Lakatos⁸⁹, MD; Grzegorz Marek Karwacki⁹⁰, MD; Paolo Candelaresi⁹¹, MD; Vincenzo Andreone⁹², MD; Carlo Maurea⁹³, MD; Roberto Tarletti, MD; Angelica Mele⁹⁴, MD; Antonio Ciacchiarelli⁹⁵, MD; Michele Alessiani⁹⁶, MD; Gabriella Monteforte⁹⁷, MD; David Pakizer⁹⁸, MSc, PhD; Martin Roubec⁹⁹, MD, PhD; David Školoudík, MD, PhD; Fenne Vandervorst¹⁰⁰, MD; Sylvie De Raedt¹⁰¹, MD, PhD; Martijn Verdam¹⁰², MD; Josef Bartoš, MD; Martin Šrámek¹⁰³, MD; Valentina Mazzoleni¹⁰⁴, MD; Luca Quilici¹⁰⁵, MD; Dario Alimonti¹⁰⁶, MD, PhD; Jessica Moller¹⁰⁷, MD; Ilaria Maestrini¹⁰⁸, MD, PhD; Marina Diomedì¹⁰⁹, MD, PhD; Giordano Lacidogna¹¹⁰, MD, PhD; Osama O. Zaidat¹¹¹, MD, MS; Eugene Lin¹¹², MD; Mohammad AlMajali¹¹³, MD; Claudio Baracchini¹¹⁴, MD; Matteo Zaccagnino¹¹⁵, MD; Federica Viaro¹¹⁶, MD; Alessandro Pezzini¹¹⁷, MD; Giulia Avola¹¹⁸, MD; Chiara Ferraro¹¹⁹, MD; Milena Świtońska¹²⁰, MD, PhD; Paulina Sobieszak-Skura, MD, PhD; Grzegorz Meder¹²¹, MD, PhD; Lukas Mayer-Suess¹²², MD, PhD; Michael Knoflach¹²³, MD; Elke Ruth Gizewski¹²⁴, MD; Matthew R Common¹²⁵, MB, BCh, BAO, LRCSI, BSc Eng, MSc; Patrick Nicholson¹²⁶, MB, BCh, BAO, MRCSI, FFR(RCSI); Sarah Power, FFR(RCSI); Marta Małgorzata Nowakowska-Kotas, MD, PhD; Michał Pula¹²⁷, MD; Maciej Guzinski¹²⁸, MD, PhD; Soledad Pérez Sánchez¹²⁹, MD, PhD; Joan Montaner¹³⁰, MD, PhD; Alexandra Sevilla Bravo¹³¹, MD; Simona Marcheselli¹³², MD; Francesca Vodret, MD; Costanza Rapillo¹³³, MD; Liliana Pereira¹³⁴, MSc; Miguel Rodrigues¹³⁵, MSc; Adam Jaros¹³⁶, MD, MPhil; Martin Kovar¹³⁷, MD; Jan Vojík, MD; Gloria Valcamonica¹³⁸, MD; Serena Gallo Cassarino, MD; Alessandra Cardillo, MD;

Correspondence to: Ghil Schwarz, MD, Department of Neurology and Stroke Unit, ASST Grande Ospedale Metropolitano Niguarda - Piazza Ospedale Maggiore 3 (20162), Milan, Italy. Email: ghilblack@gmail.com

This manuscript was sent to Harold P. Adams, Jr., Guest Editor, for review by expert referees, editorial decision, and final disposition.

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.126.056847>.

© 2026 American Heart Association, Inc.

Stroke is available at www.ahajournals.org/journal/str

Marco Petruzzellis¹, MD, PhD; Silvia Grimaldi², MD; Nicola Marrone³, MD; Ameer E. Hassan⁴, DO; Samantha Miller⁵, MD; Mohammad W. Khasawneh⁶, MD; Valentina Poretto⁷, MD; Simone Tonello, MD; Saverio Tollo, MD; Aleksander Dębiec, MD; Jacek Staszewski⁸, MD, PhD; Adam Stepień⁹, MD, PhD; Francisco Bernardo¹⁰, MD; Jorge Ferrao¹¹, MD; Marialuisa Zedde¹², MD; Rosario Pascarella¹³, MD; Alessandra Sanna¹⁴, MD, PhD; Rossella Meloni¹⁵, MD; Kati Delibas¹⁶, MD; Simona Sacco¹⁷, MD; Maria Grazia Vittorini¹⁸, MD; Raffaele Ornello¹⁹, MD, PhD; Marina Mannino²⁰, MD; Valeria Terruso²¹, MD; Marco Filizzolo, MD; Emily Hall²², MBChB; Christine Roffe²³, MD; Małgorzata Dorobek²⁴, MD, PhD; Justyna Zielinska-Turek²⁵, MD, PhD; Giuseppe Scopelliti²⁶, MD; Leonardo Pantoni²⁷, MD, PhD; Guglielmo Carlo Pero²⁸, MD; Antonio Macera²⁹, MD; Amedeo Cervo³⁰, MD; Simone Bellavia, MD; Marco Longoni³¹, MD; Mine Sezgin³², MD; Yavuz Dilmen, MD; Esme Ekizoglu³³, MD; Georgios Tsivgoulis³⁴, MD, PhD; Aikaterini Theodorou³⁵, MD, PhD; Vlad Eugen Tiu, MD, PhD; Cristina Aura Panea³⁶, MD, PhD; Simona Petrescu³⁷, MD, PhD; Mariangela Piano³⁸, MD; Thanh N. Nguyen³⁹, MD; Maria Sessa⁴⁰, MD

BACKGROUND: Contrast-associated acute kidney injury (CA-AKI) after endovascular thrombectomy (EVT) worsens outcomes in acute ischemic stroke, but threshold-based definitions do not capture prognostic information from subthreshold changes. We tested whether early creatinine change within 48 hours of EVT conveys graded prognostic information beyond dichotomous CA-AKI.

METHODS: We performed a secondary analysis of the CAN-REST (Contrast-Associated Nephropathy After Endovascular Stroke Treatment; REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06596603) registry, a retrospective multicenter observational cohort study of 6638 EVT-treated adults with acute ischemic stroke enrolled at 73 centers in 16 countries between January 1 and December 31, 2023. Early creatinine change was quantified as the natural logarithm ratio of 48-hour creatinine to baseline creatinine ($\ln[48\text{-hour/baseline creatinine}]$), with CA-AKI defined per Kidney Disease: Improving Global Outcomes criteria (≥ 0.3 mg/dL or $\geq 1.5\times$ baseline rise within 48 hours). Multivariable logistic regression adjusted for prespecified baseline and periprocedural covariates examined associations with 90-day modified Rankin Scale shift and in-hospital mortality; linear versus restricted cubic spline specification was selected by likelihood ratio testing, and contrasts were estimated at $\pm 10\%$, $\pm 25\%$, and $\pm 50\%$.

RESULTS: Greater creatinine worsening within 48 hours of EVT was independently associated with an unfavorable 90-day modified Rankin Scale shift (adjusted common odds ratio per 1-unit log-ratio increase, 1.48 [95% CI, 1.07–2.04]; linear) and in-hospital mortality (nonlinear dose-response, of larger magnitude). A 10% increase in creatinine at 48 hours—well below Kidney Disease: Improving Global Outcomes threshold—was associated with worse functional outcome (adjusted common odds ratio, 1.04 [95% CI, 1.01–1.07]) and death (adjusted odds ratio, 1.22 [95% CI, 1.13–1.32]); a 10% decrease in creatinine at 48 hours was associated with lower odds of both outcomes (adjusted common odds ratio, 0.96 [95% CI, 0.93–0.99] and 0.74 [95% CI, 0.65–0.86]). The functional-outcome association concentrated at upper modified Rankin Scale cut-points (severe disability, death); effect modification by baseline renal function was not significant.

CONCLUSIONS: Early creatinine change after EVT conveys graded, outcome-specific prognostic information extending below formal CA-AKI thresholds, with a dose-response detectable at changes as modest as $\pm 10\%$, complementing binary CA-AKI as a framework for post-EVT renal risk assessment.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06596603.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: acute kidney injury ■ creatinine ■ endovascular procedures ■ prognosis ■ stroke

Endovascular thrombectomy (EVT) has transformed the management of acute ischemic stroke due to large-vessel occlusion, yielding substantial reductions in disability and death across a broad spectrum of patients.^{1,2} As treatment eligibility has expanded³ and procedural complexity has grown, there is increasing recognition that periprocedural factors beyond the neurological injury itself may meaningfully influence clinical trajectories. Among these, renal function has attracted growing attention. A robust body of evidence supports bidirectional interactions between acute kidney injury

and acute brain injury, mediated by shared pathophysiological mechanisms including systemic inflammation, hemodynamic instability, and neurohormonal dysregulation.⁴ In the EVT setting, this brain–kidney crosstalk has particular clinical relevance: patients are simultaneously exposed to iodinated contrast media, procedural hemodynamic stress, and the systemic sequelae of acute ischemia—all of which may perturb renal function in the acute and early postprocedural period.

Despite this growing interest, the assessment of renal function in acute stroke research remains largely

Nonstandard Abbreviations and Acronyms	
CA-AKI	contrast-associated acute kidney injury
eGFR	estimated glomerular filtration rate
EVT	endovascular thrombectomy
IQR	interquartile range
KDIGO	Kidney Disease: Improving Global Outcomes
mRS	modified Rankin Scale

static. Studies consistently characterize kidney function at the time of hospital admission, while its postprocedural trajectory—arguably more informative of the physiological response to the ischemic insult and its treatment—has received comparatively little attention. Contrast-associated acute kidney injury (CA-AKI), defined by the Kidney Disease: Improving Global Outcomes (KDIGO) criteria as a serum creatinine increase of ≥ 0.3 mg/dL or ≥ 1.5 -fold above baseline within 48 hours of contrast exposure,⁵ has emerged as an independent determinant of worse short- and long-term outcomes after EVT.^{6–8} Threshold-based definitions of CA-AKI serve an essential role in clinical practice and epidemiological research, providing standardized, reproducible, and actionable criteria for risk identification. However, they also collapse a continuous biological process into a binary classification, leaving potentially relevant prognostic information uncharacterized across the full spectrum of creatinine change—including subthreshold increases that do not fulfill formal AKI criteria and postprocedural creatinine decreases, a largely unexplored phenomenon in the EVT setting.⁹ Whether early creatinine change across its full range reflects graded risk processes, rather than a purely threshold-driven relationship, remains unclear.

Against this background, we conducted a secondary analysis of the international CAN-REST (Contrast-Associated Nephropathy After Endovascular Stroke Treatment) registry to test whether early creatinine change within 48 hours of EVT—modeled as a continuous exposure across its full range, including both subthreshold increases and decreases—provides independent prognostic information on functional recovery and survival beyond what is captured by dichotomous CA-AKI classification.

METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request, subject to approval by the CAN-REST steering committee and to the data use agreements in place with the participating centers. The analytic code (Stata) is available from the corresponding author on reasonable request.

Study Design and Data Source

This study is a secondary analysis of the CAN-REST (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06596603) registry,⁶ an international multicenter observational study designed to characterize the incidence and prognostic implications of contrast-associated acute kidney injury in patients undergoing EVT for acute ischemic stroke. CAN-REST enrolled consecutive adult patients across 73 academic and community stroke centers in 16 countries between January 1 and December 31, 2023. The present analysis included patients with available baseline and 48-hour creatinine measurements, no end-stage kidney disease requiring dialysis at baseline, and complete 90-day modified Rankin Scale follow-up. We examined the association between early postprocedural creatinine change and clinical outcomes. This study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guideline for observational studies.

Creatinine Dynamics and Outcomes

Early creatinine change was quantified as the natural logarithm of the ratio between creatinine measured within 48 hours after the procedure and baseline creatinine ($\ln[48\text{-hour/baseline creatinine}]$). This formulation captures relative proportional changes symmetrically on the log scale and allows effect estimates to be expressed on a clinically interpretable multiplicative scale. The estimated glomerular filtration rate (eGFR) was calculated using the Chronic Kidney Disease Epidemiology Collaboration equation¹⁰ at both time points (baseline and 48 hours).

Two clinical outcomes were evaluated. In-hospital mortality was defined as death occurring during the index hospitalization. Functional outcome at 90 days was assessed using the modified Rankin Scale (mRS) and analyzed as an ordinal shift across the full score range (0 indicating no symptoms to 6 indicating death). The 90-day mRS was ascertained at each participating center by a stroke neurologist or by trained study personnel, either at a face-to-face follow-up visit or through a structured telephone interview with the patient or a proxy, according to local standard poststroke follow-up practice, and was subsequently abstracted into the registry. Assessments were not centrally adjudicated and were not blinded to periprocedural or laboratory data.

Statistical Analysis

Categorical variables are reported as counts and percentages, and continuous variables as medians with interquartile ranges. Missing data were handled using multiple imputation with 10 imputations, and estimates were pooled using Rubin rules.¹¹

Primary Analyses

Two multivariable adjustment models were prespecified. The primary model (Model 1) included baseline and early periprocedural covariates selected a priori based on biological plausibility and prior evidence⁶: age, sex, prestroke mRS, diabetes, atrial fibrillation, active cancer, baseline National Institutes of Health Stroke Scale score, admission glycemia, hemoglobin, log-transformed baseline creatinine, intravenous thrombolysis, anterior versus posterior circulation stroke, Alberta Stroke Program Early Computed Tomography Score (or Posterior Circulation

Alberta Stroke Program Early Computed Tomography Score, as appropriate), onset-to-groin time, successful reperfusion (modified Thrombolysis in Cerebral Infarction $\geq 2b$), early neurological deterioration, and other potential non-contrast-related causes of acute kidney injury. Early neurological deterioration was defined as an increase of ≥ 4 National Institutes of Health Stroke Scale points at 24 hours after EVT in the absence of hemorrhagic transformation. Other potential causes of acute kidney injury included hemodynamic instability, urinary obstruction, nephrotoxic drug exposure, primary renal disorders, autoimmune conditions, infections, systemic inflammation, and thrombotic or vascular complications; in the absence of standardized diagnostic criteria, these were adjudicated by the treating neurologist at each center and considered present if occurring between symptom onset and the 48-hour creatinine measurement.

The extended model (Model 2) additionally adjusted for symptomatic intracranial hemorrhage, defined as any parenchymal hemorrhage associated with a ≥ 4 -point worsening in National Institutes of Health Stroke Scale. Because symptomatic intracranial hemorrhage may represent a downstream consequence of the stroke rather than an independent baseline characteristic, Model 2 was used as a robustness analysis.

Associations with 90-day functional outcome were modeled using ordinal logistic regression across the full mRS range; in-hospital mortality was analyzed using binary logistic regression. All models accounted for the nonindependence of observations within centers through robust standard errors clustered at the center level.¹²

Nonlinearity and Effect Size Interpretation

Potential nonlinear associations between early creatinine change and each outcome were formally evaluated by comparing a restricted cubic spline model (4 knots at the 5th, 35th, 65th, and 95th percentiles¹³ of the exposure distribution) with a linear specification using a likelihood ratio test on complete-case data. The model providing the best fit for each outcome was selected based on this test: a linear model was used when linearity could not be rejected, and a restricted cubic spline model was used when nonlinearity was detected. To facilitate clinical interpretation, model-based contrasts were estimated at creatinine changes of $\pm 10\%$, $\pm 25\%$, and $\pm 50\%$ relative to baseline using standard nonlinear transformation of model coefficients and associated standard errors. These thresholds were chosen to render the continuous creatinine-change exposure in clinically interpretable terms, spanning the relevant range of early change: $\pm 50\%$ corresponds to the KDIGO relative threshold for CA-AKI (a 1.5-fold change in creatinine), while $\pm 10\%$ and $\pm 25\%$ test whether prognostic information extends below this threshold, into the subthreshold range not captured by binary CA-AKI classification. These are model-based estimates from the continuous exposure, not patient categories. The KDIGO definition of CA-AKI is met by either a $\geq 50\%$ relative increase in creatinine or an absolute increase ≥ 0.30 mg/dL within 48 hours; the overlap between the 2 thresholds depends on baseline creatinine: when baseline is >0.60 mg/dL, a relative rise $<50\%$ may still meet the absolute threshold of 0.30 mg/dL, whereas at baseline values ≤ 0.60 mg/dL, a 0.30 mg/dL increase already corresponds to a $\geq 50\%$ relative rise (since 0.30 mg/dL equals 50% of 0.60 mg/dL). To quantify this dependency, a prespecified descriptive analysis tabulated the

proportion of patients meeting the KDIGO CA-AKI definition within each relative-change stratum ($\pm 10\%$, $\pm 25\%$, $\pm 50\%$).

Proportional Odds Assumption

The proportional odds assumption underlying the ordinal logistic regression model was formally evaluated using a likelihood ratio test comparing a partial proportional odds model—in which the effect of the creatinine ratio was allowed to vary freely across mRS cut-points while all adjustment covariates were constrained to a proportional odds structure—against the fully constrained proportional odds model. Cut-point-specific estimates from the partial proportional odds model are reported as a descriptive analysis of how the association is distributed across the mRS spectrum; the adjusted common odds ratio from the proportional odds model was retained as the primary summary estimate.

Secondary Analyses and Effect Modification

The association between creatinine change and outcomes was examined for potential effect modification by baseline renal function. The primary approach consisted of a multiplicative interaction term between creatinine dynamics and baseline eGFR stratum (<60 versus ≥ 60 mL/min per 1.73 m²) included in the primary model, with stratum-specific estimates and *P* value for interaction reported; effect modification was considered statistically significant at $P < 0.10$.

Sensitivity Analyses

A prespecified sensitivity analysis excluded patients with extreme creatinine changes, defined as an absolute log-creatinine ratio exceeding 1.0—corresponding to a >2.7 -fold increase or decrease from baseline—to assess robustness of primary findings to potential measurement error or physiologically nonrepresentative fluctuations.

All statistical tests were 2-sided, and $P < 0.05$ was considered statistically significant. Analyses were performed using Stata, version 18 (StataCorp, College Station, TX). Figures were produced in Stata from the analytic output; a generative AI tool (Claude, Anthropic) was used only for cosmetic refinement of the Stata-generated plots (axis labels, fonts, color palette, legend layout). The tool did not access the study data or alter any plotted geometry; figures were visually verified against the original Stata output.

RESULTS

Study Population

Between January 1 and December 31, 2023, 8871 patients underwent EVT for acute ischemic stroke at the 73 participating centers. Of these, 2233 (25.2%) were excluded: 5 ($<0.1\%$) were younger than 18 years, 21 (0.2%) had end-stage kidney disease requiring dialysis, 1414 (15.9%) lacked a baseline and a 48-hour creatinine measurement, and 793 (8.9%) had no 90-day mRS assessment. The analytic cohort therefore comprised 6638 patients, representing 74.8% of all EVT-treated patients at participating centers during the study period, and corresponded to the population of the primary CAN-REST publication⁶ (Figure S1). Baseline characteristics

of the analytic cohort have been described in detail in our previous publication.⁶ Briefly, the median age was 74 years (interquartile range [IQR], 64–82) and 48.7% were male. The median baseline National Institutes of Health Stroke Scale score was 15 (IQR, 10–20), and 90.2% of strokes involved the anterior circulation. Successful reperfusion (modified Thrombolysis in Cerebral Infarction $\geq 2b$) was achieved in 89.4% of patients. Median baseline creatinine was 0.92 mg/dL (IQR, 0.76–1.16), corresponding to a median eGFR of 77 mL/min per 1.73 m² (IQR, 57–92). The median creatinine measured within 48 hours of the procedure was 0.88 mg/dL (IQR, 0.71–1.10). In-hospital mortality occurred in 902 (13.6%) patients, and the median 90-day mRS score was 3 (IQR, 1–5); severe disability or death at 90 days (mRS ≥ 4) was observed in 2778 (41.9%) patients. Baseline characteristics are shown in Table 1.

Distribution of Early Creatinine Change

The distribution of early creatinine change across the study population is shown in Table S1 and illustrated graphically in Figure 1. Most patients showed either a modest decrease (40.1% with a reduction between 5% and 25%) or no meaningful change (27.8%, within $\pm 5\%$ of baseline). Creatinine worsening was observed in 23.7% of patients: increases between 5% and $<10\%$ in 6.5% ($n=433$), between 10% and $<25\%$ in 10.8% ($n=717$), between 25% and $<50\%$ in 4.1% ($n=275$), and $\geq 50\%$ —corresponding to the KDIGO relative threshold for CA-AKI—in 2.2% ($n=146$). A marked creatinine decrease exceeding 25% was observed in 8.5% of patients, including 30 (0.5%) with a reduction exceeding 50%. Applying the full KDIGO CA-AKI definition (relative or absolute criterion) classified 4.9% of the cohort ($n=326$) as CA-AKI, of whom 55.2% ($n=180$) were identified exclusively through the absolute criterion (≥ 0.30 mg/dL). Within the relative-increase bands lying below the $\geq 50\%$ relative threshold, the KDIGO absolute criterion was met by 5.3% (38/717) of patients with a $+10\%$ to $+25\%$ rise and by 49.5% (136/275) of those with a $+25\%$ to $+50\%$ rise (Table S2).

Association Between Early Creatinine Change and Clinical Outcomes

Formal testing of the linearity assumption supported a linear model for 90-day mRS shift (likelihood ratio test $\chi^2[2]=4.59$, $P=0.101$; Figure 2A) and a restricted cubic spline model for in-hospital mortality (likelihood ratio test $\chi^2[2]=13.17$, $P=0.001$; Figure 2B). No single summary odds ratio is reported for mortality; the full dose-response relationship is characterized through prespecified contrasts (Table 2).

In the primary multivariable model (Model 1), each 1-unit increase in the log-creatinine ratio was

independently associated with higher odds of an unfavorable ordinal shift across the 90-day mRS, summarized by an adjusted common odds ratio of 1.48 (95% CI, 1.07–2.04). Results were consistent after additional adjustment for symptomatic intracranial hemorrhage in Model 2 (mRS shift: adjusted common OR, 1.40 [95% CI, 1.02–1.92]).

Prespecified contrasts across the creatinine-change continuum demonstrated a graded dose-response pattern for both outcomes (Table 2; Figure 2). For the 90-day mRS shift, creatinine worsening was associated with an unfavorable ordinal shift at all prespecified contrasts: a 10% increase corresponded to an adjusted common OR of 1.04 (95% CI, 1.01–1.07), a 25% increase to an adjusted common OR of 1.09 (95% CI, 1.01–1.17), and a 50% increase to an adjusted common OR of 1.17 (95% CI, 1.02–1.32). Creatinine decreases were associated with progressively lower odds of worse functional outcome (10% decrease: adjusted common OR, 0.96 [95% CI, 0.93–0.99]; 25% decrease: adjusted common OR, 0.89 [95% CI, 0.81–0.98]; 50% decrease: adjusted common OR, 0.76 [95% CI, 0.59–0.93]). For in-hospital mortality, associations were of larger magnitude: a 10% creatinine increase was associated with an adjusted OR of 1.22 (95% CI, 1.13–1.32), a 25% increase with an adjusted OR of 1.42 (95% CI, 1.24–1.63), and a 50% increase with an adjusted OR of 1.67 (95% CI, 1.30–2.14). Creatinine decreases of 10% and 25% were associated with lower odds of in-hospital mortality (adjusted OR, 0.74 [95% CI, 0.65–0.86] and adjusted OR, 0.68 [95% CI, 0.55–0.82], respectively); a 50% decrease was not significantly associated with mortality risk (adjusted OR, 0.90 [95% CI, 0.44–1.83]).

Proportional Odds Assumption

Formal evaluation of the proportional odds assumption revealed that the effect of creatinine change on 90-day mRS was not uniform across the outcome spectrum (likelihood ratio $\chi^2=28.53$, 5 df, $P<0.001$; Table S3). The association was negligible and nonsignificant at the lower mRS cut-points (mRS 0 versus 1–6 through mRS 0–3 versus 4–6; all $P>0.05$), and large and statistically significant at the upper cut-points (mRS 0–4 versus 5–6: OR, 2.18 [95% CI, 1.45–3.27]; mRS 0–5 versus 6: OR, 2.22 [95% CI, 1.47–3.37]; both $P<0.001$), indicating that the association with creatinine dynamics is concentrated at the boundary between severe disability and death.

Effect Modification by Baseline Renal Function

The association between early creatinine change and outcomes did not differ significantly by baseline renal function status. The P value for the multiplicative interaction between creatinine dynamics and eGFR

Table 1. Baseline Characteristics, Clinical Variables, and Outcomes in the CAN-REST Population

	N=6638	Missing, N (%)
Demographics and baseline characteristics		
Age, y; median (IQR)	74 (64–82)	0 (0)
Male sex	3231 (48.7)	0 (0)
White ethnicity	5745 (92.8)	446 (6.7)
BMI, median (IQR)	27 (24–30)	1904 (28.7)
Prestroke mRS, median (IQR)	0 (0–1)	82 (1.2)
Medical history and risk factors		
Known chronic kidney disease	2604 (39.2)	0 (0)
Arterial hypertension	4876 (73.6)	9 (0.1)
Diabetes	1727 (26.0)	7 (0.1)
Dyslipidemia	3318 (50.1)	14 (0.2)
Atrial fibrillation	2355 (35.6)	15 (0.2)
Coronary artery disease and heart failure	1872 (28.3)	18 (0.3)
Active cancer	493 (7.7)	234 (3.5)
Antiplatelet therapy	1835 (27.6)	0 (0)
Anticoagulant therapy	1464 (22.1)	0 (0)
RAAS inhibitors	2742 (42.6)	206 (3.1)
Metformin	844 (14.5)	811 (12.2)
NSAIDs within 48 h before EVT	331 (5.6)	711 (10.7)
Acute-phase variables		
Baseline NIHSS score, median (IQR)	15 (10–20)	63 (0.9)
Mean arterial pressure, mm Hg; median (IQR)	103 (93–115)	291 (4.4)
Admission blood glucose, mg/dL; median (IQR)	123 (106–151)	189 (2.8)
Admission hemoglobin, g/dL; median (IQR)	13.3 (12.0–14.4)	58 (0.9)
Baseline creatinine, mg/dL; median (IQR)	0.92 (0.76–1.16)	0 (0)
Baseline eGFR, mL/min per 1.73 m ² ; median (IQR)	77 (57–92)	0 (0)
Intravenous thrombolysis	2879 (43.4)	1 (0.0)
Baseline imaging variables		
Pre-EVT contrast volume, mL; median (IQR)	90 (70–100)	1609 (24.2)
Large-vessel occlusion (vs medium vessel occlusion)	4856 (74.1)	85 (1.3)
Tandem lesion	960 (14.6)	68 (1.0)
Anterior circulation (vs posterior)	5910 (90.2)	86 (1.3)
ASPECTS/pc-ASPECTS, median (IQR)	9 (8–10)	514 (7.7)
EVT-related variables		
Onset to EVT, min; median (IQR)	255 (175–410)	254 (3.8)
EVT procedure duration, min; median (IQR)	40 (23–62)	644 (9.7)
No. of EVT passes, median (IQR)	1 (1–3)	252 (3.8)
EVT contrast volume, mL; median (IQR)	80 (50–120)	1890 (28.5)
Procedural hypotension	613 (12.2)	1620 (24.4)
Successful reperfusion (mTICI ≥2b)	5731 (89.4)	227 (3.4)
Early post-EVT variables		
Additional contrast exposure (<48 h post-EVT)	700 (11.6)	579 (8.7)
Early neurological deterioration	400 (6.9)	235 (3.5)
Symptomatic hemorrhagic transformation	351 (5.5)	235 (3.5)
Peri-EVT IV hydration <500 mL	697 (12.5)	1076 (16.2)
Peri-EVT IV hydration 500–1000 mL	2305 (41.4)	
Peri-EVT IV hydration >1000 mL	2560 (46.0)	

(Continued)

Table 1. Continued

	N=6638	Missing, N (%)
Other potential causes of acute kidney injury	1108 (17.7)	385 (5.8)
Creatinine within 48 h, mg/dL; median (IQR)	0.88 (0.71–1.10)	0 (0)
Outcomes		
Length of hospital stay, d; median (IQR)	8 (4–13)	–
In-hospital mortality	902 (13.6)	26 (0.4)
90-d mRS score, median (IQR)	3 (1–5)	0 (0)
Severe disability or death at 90 d (mRS ≥4)	2778 (41.9)	0 (0)

Data are presented as median (IQR) for continuous variables and N (%) for categorical variables. Missing values are reported as N (%). Known chronic kidney disease includes all patients with a documented history of chronic kidney disease of any stage at the time of admission. ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; BMI, body mass index; CAN-REST, Contrast-Associated Nephropathy After Endovascular Stroke Treatment; eGFR, estimated glomerular filtration rate; EVT, endovascular thrombectomy; IQR, interquartile range; mRS, modified Rankin Scale; mTICI, modified Thrombolysis in Cerebral Infarction; NIHSS, National Institutes of Health Stroke Scale; NSAID, nonsteroidal anti-inflammatory drug; pc-ASPECTS, Posterior Circulation Alberta Stroke Program Early Computed Tomography Score; and RAAS, renin-angiotensin-aldosterone system.

stratum (<60 versus ≥60 mL/min per 1.73 m²) was 0.855 for 90-day mRS shift and 0.346 for in-hospital mortality.

Sensitivity Analysis

Results were robust after excluding 21 patients with extreme creatinine changes (absolute log-creatinine ratio >1, corresponding to a >2.7-fold change from baseline). The association between creatinine change and 90-day mRS shift remained consistent in both Model 1 (adjusted common OR, 1.62 [95% CI, 1.12–2.34]) and Model 2 (adjusted common OR, 1.52 [95% CI, 1.06–2.19]). For in-hospital mortality, the graded dose-response pattern was preserved with magnitudes closely matching the primary analysis: in Model 1, the adjusted OR was 1.26 (95% CI, 1.17–1.36) at a +10% relative rise, 1.54 (1.33–1.78) at +25%, and 1.96 (1.47–2.61) at +50%, with concordant estimates in Model 2 (+10%: 1.25 [1.16–1.35]; +25%: 1.47 [1.26–1.71]; +50%: 1.73 [1.27–2.36]); all worsening contrasts *P*<0.001.

DISCUSSION

In this large international cohort of 6638 patients undergoing EVT for acute ischemic stroke, early creatinine change within 48 hours conveyed graded and independent prognostic information on both functional recovery and survival, with associations extending well below formal CA-AKI thresholds. Four findings are central. (1) The relationship with clinical outcomes is continuous, with incremental risk detectable at creatinine increases as modest as 10%. (2) The shape of this association differs by outcome: nonlinear and larger for in-hospital mortality, linear and attenuated for 90-day disability. (3) The functional-outcome effect is concentrated at the upper end of the modified Rankin Scale, indicating that creatinine worsening tracks fatal or near-fatal events rather than the graded spectrum

of neurological recovery. (4) The signal is symmetrical: early creatinine decreases were independently associated with better functional outcome and lower mortality, positioning postprocedural renal improvement as an informative marker rather than merely the absence of injury.

These findings extend an emerging body of evidence linking postprocedural renal function to outcomes after EVT. Prior work from the CAN-REST registry and other multicenter cohorts has established that formally defined CA-AKI—occurring in ≈5% of EVT-treated patients^{6,7}—is an independent predictor of worse short- and long-term outcomes after adjustment for established neurological prognostic factors. Consistent with this prior work, patients meeting the formal KDIGO threshold remained those at highest risk. The present analysis advances this line of inquiry by demonstrating that the prognostic signal, while most pronounced in this group, is not confined to it and extends continuously across the entire creatinine-change spectrum. A 10% creatinine increase—well below the 1.5-fold KDIGO relative threshold—was associated with 22% higher odds of in-hospital death and 4% higher odds of unfavorable functional shift. Threshold-based classification, while operationally convenient and clinically essential, leaves additional prognostic information uncharacterized in the majority of patients who do not fulfill formal CA-AKI criteria but nonetheless experience subclinical renal stress. It should be noted that the KDIGO definition of CA-AKI combines 2 criteria—a relative increase ≥1.5-fold and an absolute increase ≥0.30 mg/dL within 48 hours—so that patients whose relative rise falls below the +50% threshold may still meet the absolute criterion. This partial overlap is non-negligible in the +25% to <+50% band, where 136 of 275 patients (49.5%) also met the absolute criterion. In contrast, the subthreshold nature of the signal is clearly illustrated by the +10% contrast: of the 717 patients with a creatinine rise between +10% and <+25%, only 38 (5.3%) met the KDIGO absolute criterion of ≥0.30 mg/dL, suggesting

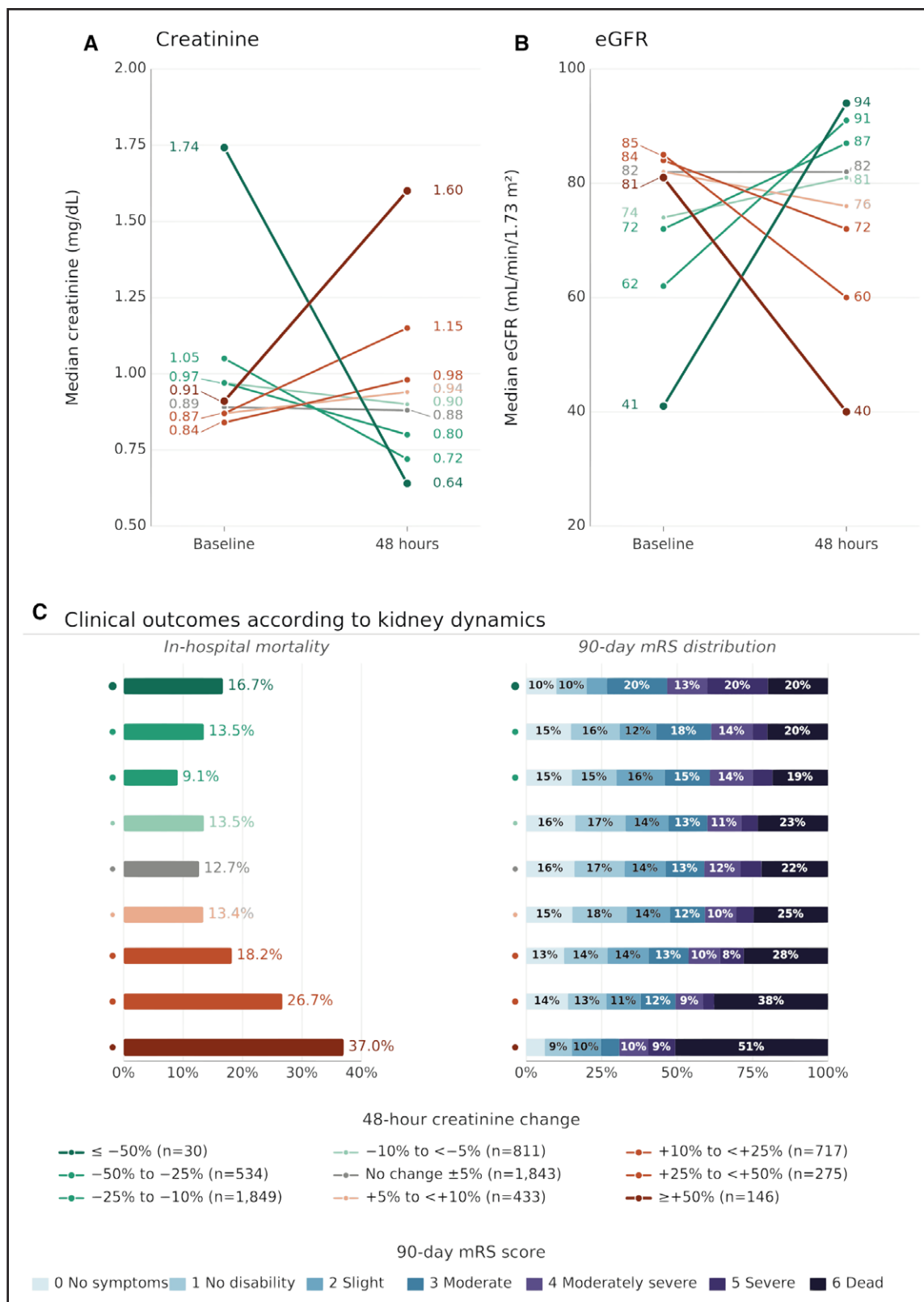


Figure 1. Physiological trajectories and clinical outcomes across 48-hour creatinine-change bands after endovascular thrombectomy.

A, Median serum creatinine (mg/dL) at baseline and 48 h per band. **B**, Median estimated glomerular filtration rate (eGFR; mL/min per 1.73 m², Chronic Kidney Disease Epidemiology Collaboration 2021) at the same time points. **C**, Clinical outcomes per band: (**C1**) in-hospital mortality (%); (**C2**) 90-day modified Rankin Scale (mRS) distribution as stacked bars.

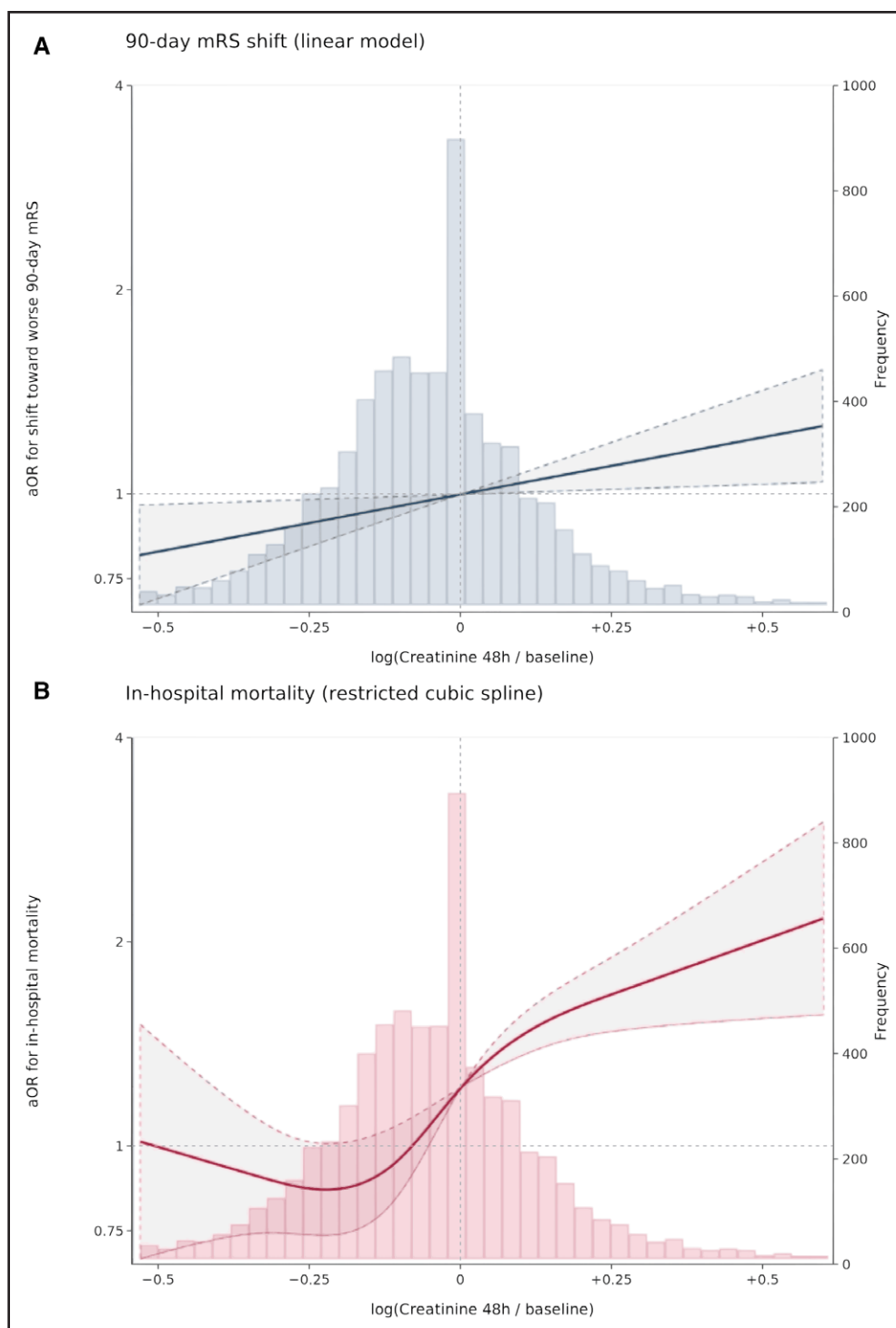


Figure 2. Association between early creatinine change and clinical outcomes after endovascular thrombectomy.

A, Adjusted common odds ratio (acOR) for a shift toward worse 90-day functional outcome (ordinal modified Rankin Scale [mRS]), from a linear ordinal logistic model. **B**, Adjusted odds ratio (aOR) for in-hospital mortality, from a restricted cubic spline logistic model (4 knots positioned at the 5th, 35th, 65th, and 95th percentiles of the exposure distribution). Both models were adjusted for age, sex, prestroke mRS, diabetes, atrial fibrillation, active cancer, baseline National Institutes of Health Stroke Scale, admission glycemia, hemoglobin, log-transformed baseline creatinine, intravenous thrombolysis, anterior vs posterior circulation, Alberta Stroke Program Early Computed Tomography Score, onset-to-groin time, successful reperfusion (modified Thrombolysis in Cerebral Infarction $\geq 2b$), early neurological deterioration, and other potential concomitant causes of acute kidney injury.

that the prognostic association at this level of change is largely independent of the formal CA-AKI definition. (Table S2).

The differential shape and magnitude of the associations for the 2 outcomes merit careful consideration. The linear relationship between creatinine change and

Table 2. Association Between Early Creatinine Change and Clinical Outcomes After Endovascular Thrombectomy

Creatinine change at 48 h	90-day mRS shift*				In-hospital mortality†			
	Model 1		Model 2		Model 1		Model 2	
	aOR (95% CI)	Pvalue	aOR (95% CI)	P value	aOR (95% CI)	Pvalue	aOR (95% CI)	Pvalue
Overall association (per 1-unit increase in log[Cr ₄₈ h/baseline])								
Overall cohort (N=6638)	1.48 (1.07–2.04)	0.017	1.40 (1.02–1.92)	0.038	...‡	...	...‡	...
Prespecified contrasts relative to no creatinine change§								
	aOR (95% CI)		aOR (95% CI)		aOR (95% CI)		aOR (95% CI)	
50% Improvement	0.76 (0.59–0.93)		0.79 (0.62–0.97)		0.90 (0.44–1.83)		0.91 (0.46–1.84)	
25% Improvement	0.89 (0.81–0.98)		0.91 (0.83–0.99)		0.68 (0.55–0.82)		0.67 (0.55–0.81)	
10% Improvement	0.96 (0.93–0.99)		0.97 (0.93–1.00)		0.74 (0.65–0.86)		0.74 (0.64–0.85)	
No change (reference)	1.00 (reference)		1.00 (reference)		1.00 (reference)		1.00 (reference)	
10% Worsening	1.04 (1.01–1.07)		1.03 (1.00–1.06)		1.22 (1.13–1.32)		1.22 (1.13–1.32)	
25% Worsening	1.09 (1.01–1.17)		1.08 (1.00–1.15)		1.42 (1.24–1.63)		1.39 (1.21–1.60)	
50% Worsening	1.17 (1.02–1.32)		1.15 (1.00–1.29)		1.67 (1.30–2.14)		1.58 (1.23–2.05)	

Estimates from multiply imputed data (m=10 imputations), clustered robust standard errors (73 centers). Model 1 adjusted for: baseline creatinine (log-transformed), age, sex, prestroke mRS, diabetes, atrial fibrillation, active cancer, NIHSS, glycemia, hemoglobin, IV thrombolysis, anterior/posterior circulation, ASPECTS, onset-to-groin time, successful reperfusion (mTICI ≥2b), early neurological deterioration, and potential concomitant causes of acute kidney injury. Model 2: Model 1 additionally adjusted for symptomatic intracranial hemorrhage (sICH). acOR indicates adjusted common odds ratio; aOR, adjusted odds ratio; ASPECTS, Alberta Stroke Program Early Computed Tomography Score; mRS, modified Rankin Scale; mTICI, modified Thrombolysis in Cerebral Infarction; and NIHSS, National Institutes of Health Stroke Scale.

*90-day mRS shift: ordinal logistic regression across the full modified Rankin Scale range (0–6). Overall acOR per 1-unit increase in log(Cr₄₈h/baseline); contrasts derived from the linear coefficient. Linearity not rejected (LRT $\chi^2(2)=4.59$, $P=0.101$).

†In-hospital mortality: binary logistic regression with restricted cubic spline (4 knots: p5, p35, p65, p95). Nonlinearity detected (LRT $\chi^2(2)=13.17$, $P=0.001$). Prespecified contrasts derived from spline coefficients.

‡A single summary aOR for in-hospital mortality is not reported because a nonlinear (spline) model was used. Prespecified contrasts below fully characterize the dose-response relationship for this outcome.

§Contrasts expressed as odds ratios relative to no creatinine change (reference). CI for the 50% improvement contrast for mortality are wide due to sparse data in that region.

mRS shift suggests that each incremental unit of renal function change contributes proportionally to functional risk, without a discernible threshold or acceleration zone. The nonlinear relationship with mortality—characterized by a steep increase in risk at higher values of the log-creatinine ratio—points to a qualitatively different biological process at the severe end of the spectrum, likely reflecting the convergence of contrast nephrotoxicity, hemodynamic compromise, and systemic inflammation into a clinical syndrome carrying disproportionately high mortality risk.^{14,15} The concentration of the mRS effect at cut-points ≥5 provides convergent evidence: the contribution of creatinine dynamics to functional outcome is largely mediated through its relationship with fatal complications rather than with graded neurological disability.

Several pathways may link post-EVT creatinine worsening to the observed outcomes, although the present data cannot distinguish whether creatinine change acts as a causal mediator of worse outcomes or as a prognostic marker of underlying systemic illness severity—or both. This distinction has important implications: a causal pathway would support renal-targeted interventions, whereas a marker interpretation would support the use of early creatinine dynamics as a risk-stratification tool without implying benefit from direct renal intervention. Several nonexclusive and concurrent mechanisms are plausible. Contrast-associated kidney injury involves direct tubular cytotoxicity, intrarenal vasoconstriction

with medullary hypoxia, and oxidative injury¹⁶; subclinical renal dysfunction may also activate systemic inflammatory cascades that could propagate secondary injury in the acutely ischemic brain.¹⁷ At the same time, post-procedural creatinine worsening may equally reflect or precipitate cardiorenal syndrome, hemodynamic instability, or systemic infection¹⁸—clinical conditions strongly associated with in-hospital mortality and severe disability, and only partially captured in the present registry. Disentangling a direct kidney-to-brain effect from mediation through these systemic pathways will require prospective studies with granular capture of cardiac, hemodynamic, and infectious complications.

The observation that early creatinine decreases were independently associated with improved outcomes across both domains is perhaps the most conceptually novel aspect of this analysis. Postprocedural creatinine reduction—a phenomenon that has received virtually no attention in the EVT literature—may identify patients in whom reperfusion resolves a transient prerenal state rather than patients experiencing artifactual creatinine dilution, although this interpretation remains exploratory. The bands with the largest creatinine decreases were populated by patients with the lowest baseline eGFR, and their renal function appeared to recover substantially by 48 hours: in the ≤–50% band, median eGFR rose from 41 to 94 mL/min per 1.73 m² (Table S1; Figure 1A and 1B). A shift of this magnitude is

unlikely to be explained by periprocedural fluid administration alone and would be more consistent with restoration of renal perfusion after successful recanalization, particularly in patients whose admission creatinine reflected dehydration, prestroke hemodynamic compromise, or both—though contributions from hemodilution and regression to the mean cannot be excluded on the basis of observational data.^{19,20} This pattern is most consistent with a prerenal contribution to the elevated baseline creatinine (a reversible, perfusion-related rise without intrinsic renal injury), arising from relative volume depletion at symptom onset—reduced fluid intake, dysphagia, sympathetic activation, and prehospital delay—and corrected by periprocedural intravenous hydration (exceeding 1000 mL in 46.0% of the cohort) and restoration of systemic perfusion after recanalization. Consistent with this interpretation, the strata with the greatest creatinine decreases had the highest baseline creatinine and lowest eGFR, that is, the largest margin for normalization toward true baseline. If confirmed, this would reframe early creatinine decrease as a candidate marker of systemic treatment response rather than an incidental laboratory finding, and would support its prospective evaluation alongside established markers of reperfusion success.

The absence of significant effect modification by baseline renal function is notable: patients with preexisting chronic kidney disease might have been expected to show heightened vulnerability⁸ to postprocedural creatinine perturbations, yet the prognostic signal was homogeneous across eGFR strata. The clinical implication is that post-EVT creatinine monitoring could be informative across the spectrum of baseline renal function and should not be restricted to patients with established chronic kidney disease.

Several limitations warrant acknowledgment. As a secondary analysis of an observational registry, causal inference is not possible, and residual confounding by periprocedural hypotension, incompletely recorded fluid administration, procedural complexity, or medical complications not fully captured in the adjustment model cannot be excluded. The adjudication of potential noncontrast causes of AKI was performed by treating clinicians without standardized criteria, introducing the possibility of diagnostic misclassification. A single creatinine value within 48 hours was collected per patient (the 48-hour measurement when available, otherwise the closest within the window), precluding characterization of the temporal trajectory of renal function or its recovery. More frequent monitoring in sicker patients may have biased ascertainment toward peak values in that subgroup. The wide CIs for the 50% improvement contrast for mortality reflect the relatively limited number of patients with marked creatinine decreases. Finally, the study population was predominantly of White European ethnicity, which may

limit generalizability to populations with different ethnic backgrounds, inherited renal reserve,²¹ or prevalence of cardiometabolic comorbidities that influence baseline and postprocedural renal function. External validation in multi-ethnic cohorts, including Asian, African, and other under-represented populations, will be important before these findings are generalized beyond the present setting. In addition, the analytic cohort comprised 74.8% of all EVT-treated patients at participating centers, with exclusions driven almost entirely by an unavailable 48-hour creatinine value (15.9%) or a missing 90-day mRS (8.9%). Both requirements are more likely to be met in patients who survive the early postprocedural period and who remain in the care of the treating center, so the cohort may under-represent patients with early death or early transfer; the direction and magnitude of any resulting selection bias cannot be quantified from the available data. In a direct comparison (Table S4), included and excluded patients were closely comparable in age, vascular risk-factor profile, stroke severity, baseline creatinine and rates of intravenous thrombolysis and successful reperfusion, with differences in medians of no more than 2 units for any of these variables, although some reached statistical significance given the sample size. Excluded patients had a longer onset-to-EVT time (median 300 versus 255 minutes), a shorter hospital stay (5 versus 8 days), and lower in-hospital mortality (9.2% versus 13.6%), a pattern more consistent with early discharge or interhospital transfer than with a systematically different baseline risk. The 90-day mRS was also ascertained locally, without central adjudication or blinding to periprocedural data, which may have introduced misclassification of functional outcome.

From a methodological standpoint, these results provide a concrete illustration of the cost of dichotomization in clinical research.²² By collapsing the continuous distribution of creatinine change into a binary AKI/no-AKI classification, prior studies have restricted their inferences to the tail of the distribution while leaving the prognostic information embedded in the subthreshold population uncharacterized. The continuous framework applied here—modeling creatinine change as a log-transformed ratio and deriving prespecified contrasts at clinically meaningful thresholds—offers a complementary and more information-preserving approach that can be incorporated into future risk prediction models. Clinically, the finding that the prognostic signal extends to changes as small as 10% to 25% suggests that systematic post-EVT creatinine surveillance may identify patients at elevated risk even in the absence of overt CA-AKI. Whether early identification translates into improved outcomes through nephroprotective fluid protocols,²³ avoidance of secondary nephrotoxic exposures, or targeted hemodynamic monitoring remains an open question for prospective investigation.

CONCLUSIONS

In this large international cohort, early creatinine change within 48 hours of EVT conveyed independent, graded, and outcome-specific prognostic information extending well below formal CA-AKI thresholds, with a continuous dose-response pattern detectable at changes as modest as 10%. The association was substantially larger for in-hospital mortality than for functional disability, early creatinine improvement predicted better outcomes, and the prognostic signal was consistent regardless of baseline renal function. These findings complement binary CA-AKI classification as a framework for postprocedural renal assessment in acute stroke, and support systematic integration of continuous creatinine trajectory monitoring into post-EVT risk stratification.

ARTICLE INFORMATION

Received May 21, 2026; final revision received August 26, 2026; accepted September 9, 2026.

Affiliations

Vascular Neurology and Stroke Unit (G. Schwarz, A.C.R., S.B., M.L.), Neurology Division (M. Sessa), Department of Neurology and Stroke Unit (G. Schwarz, A.C.R., S.B., M. Sessa), and Department of Neuroradiology (G.C.P., A. Macera, A. Cervo, M. Piano), ASST Grande Ospedale Metropolitano Niguarda, Milan, Italy. Department of Statistical Science, University College London, United Kingdom (G. Ambler). Department of Neurology (P.W., Agnieszka Slowik), Student Scientific Group in Cerebrovascular Diseases, Faculty of Medicine (P.W., Agnieszka Slowik), and Faculty of Medicine, Student Scientific Group in Cerebrovascular Diseases (A.B.), Jagiellonian University Medical College, Krakow, Poland. Stroke Institute, Department of Neurology, University of Pittsburgh Medical Center, PA (M.F.D., A.R.A.-B., R.G.N.). Stroke Unit, Hospital de São José, Unidade Local de Saúde de São José, Lisboa, Portugal (A.P.N., P.F.). Centro Clínico Académico de Lisboa, Portugal (A.P.N., P.F.). IRCCS Istituto delle Scienze Neurologiche di Bologna, Department of Neurology and Stroke Center, Maggiore Hospital, Italy (M. Paolucci, A.Z.). IRCCS Istituto delle Scienze Neurologiche di Bologna, Neuroradiology Unit, Maggiore Hospital, Italy (L. Simonetti). Department of Neurology, Faculty of Medicine (N.L., Z.G.) and Department of Radiology, Faculty of Medicine (J. Fedorko), P.J. Safarik University and L. Pasteur University Hospital, Košice, Slovakia. Department of Neuroradiology, AP-HP, Henri Mondor University Hospital, DMU FlixIT, Créteil, France (L. Scaria, E.K., F.F.). Department of Neurology, Inselspital, Bern University Hospital, and University of Bern, Switzerland (M. Beyeler, M. Arnold). Institute for Diagnostic and Interventional Neuroradiology, Inselspital, Bern University Hospital, and University of Bern, Switzerland (A. Mujanovic). Department of Neuroradiology, Hospital de Braga, Portugal (T.M.). School of Medicine, University of Minho, Braga, Portugal (T.M., M. Morais). Department of Neurology, Unidade Local de Saúde do Alto Minho, Viana do Castelo, Portugal (L.F.). Department of Neurology, Institute of Medical Sciences, Medical College of Rzeszow University, Poland (M.W., H.B.-P.). University of Rzeszow, Poland (P.P.). Department of Neurology and Stroke Centre, Hospital Universitario 12 de Octubre, Instituto de Investigación Hospital 12 de Octubre (i+12), Madrid, Spain (P. Calleja). Department of Neurology and Stroke Centre, Hospital Universitario 12 de Octubre, Madrid, Spain (F.O., D. Seoane). Stroke Unit, Department of Neurology, University Emergency Hospital Bucharest, Romania (N.A., R.A.R., C.T.). Department of Clinical Neurosciences, Carol Davila University of Medicine and Pharmacy, Bucharest, Romania (R.A.R., C.T.). Department of Neurology and Neurosurgery (S.A.K.), Department of Neurology (A.A.Q.), and Department of Neurosurgery (I.S.T.), Medical University of South Carolina, Charleston. Department of Neurology, Complejo Hospitalario Universitario de Albacete, Spain (O.A.-M., F.H.-F., M. Paya, J.D.M.-N.). Department of Neurology, St. Jadwiga Provincial Specialist Hospital, Opole, Poland (B.M.L.-R., K.M.). Institute of Medicine, University of Opole, Poland (B.M.L.-R., K.M.). Department of Human Neurosciences, Sapienza University of Rome, Italy (D.T., E.B.). Student Scientific Society at the Department of Neurology, Institute of Medicine, University of Opole, Poland (M.L.R.). Stroke Unit, Emergency Department, Policlinico Umberto I Hospital, Sapienza University of Rome, Italy (M.D.M.). The University of Kansas Medical Center, KS (P.E., T.B.). The University of Kansas Health System, KS (M.L.H.). McGovern Medical School,

UTHealth Houston, TX (A.A., W.M.B., S.A.S.). KU Leuven, Leuven Brain Institute, Department of Neurosciences, Experimental Neurology, Belgium (J.D., L.M., A.W.). Department of Neurology, University Hospitals Leuven, Belgium (J.D., L.M., A.W.). Department of Neurology, Hospital de Egas Moniz, Centro Hospitalar Lisboa Ocidental, Lisbon, Portugal (J.P.M., J.B.C.). UO Neurologia d'Urgenza e Stroke Unit, IRCCS Fondazione Mondino, Pavia, Italy (N.D.L.). UOC Neuroradiologia, IRCCS Policlinico San Matteo, Pavia, Italy (A.R.). UO Neurologia d'Urgenza e Stroke Unit, IRCCS Fondazione Mondino, Pavia, Italy (C.T.C.). Dipartimento di Scienze del Sistema Nervoso e del Comportamento, Università di Pavia, Italy (C.T.C.). Department of Neurology and Neurological Nursing (P.L.), Department of Neurology (M. Seweryn), and Department of Neurology (K.R.), Medical University of Lublin, Poland. Departments of Neurology and Radiology, Boston Medical Center (P.K.), Department of Radiology, Boston Medical Center (M. Abdalkader), and Department of Neurology, Boston Medical Center (T.N.N.), Boston University Chobanian and Avedisian School of Medicine, MA. Department of Neurology & Stroke Unit (V.S., T.T.) and Neuroradiology Unit (Antico Sanna), Santa Corona Hospital, Pietra Ligure (SV), Italy. Stroke Unit, Department of Medical Sciences, Surgery and Neurosciences, University of Siena, Italy (M. Acampa). U.O.C. Stroke Unit, Azienda Ospedaliera Universitaria Senese, Siena, Italy (F.R., R. Tassi). Neurological Ward with Stroke Unit, St John Paul II Western Hospital in Grodzisk Mazowiecki, Poland (M.E.B., A.M.B.-S.). Faculty of Medicine, Lazarski University, Warsaw, Poland (M.E.B., A.M.B.-S.). Department of Neurosciences, Corewell Health West and Michigan State University, Grand Rapids (J. Min, N.H.A.). UOSD Stroke Unit, S. Camillo-Forlanini Hospital, Rome, Italy (M. Mangiardi, S.A., E.P.). Department of Neurology and Stroke Centre, La Paz Institute for Health Research – IdiPAZ (La Paz University Hospital – Autonomous University of Madrid), Spain (C.H.-T., R.R., B.F.). Stroke Center, Service of Neurology, Department of Clinical Neurosciences (D. Strambo, C.M.T.) and Neuroradiology Unit, Service of Diagnostic and Interventional Radiology, Department of Medical Radiology (G. Saliou), University Hospital of Lausanne and University of Lausanne, Switzerland. A.O.E. Cannizzaro, Catania, Italy (A.S.C., D.M.). Department of Neuroscience, Catholic University of the Sacred Heart, Rome, Italy (P.A.R., M. Moci, I.S.). Department of Neurology, Hospital Universitario Ramon y Cajal, Madrid, Spain (A.C.-C., R.V.L., S.G.M.). Stroke Unit, Udine University Hospital, Italy (G. Merlino). Clinical Neurology, Udine University Hospital, Italy (G. Merlino, M. Valente, A. Todesco). DMED, Udine University Hospital, Italy (M. Valente, A. Todesco). Department of Neurology, Cantonal Hospital of Lucerne, Switzerland (M. Bolognese, L.-B.L.). Department of Radiology and Nuclear Medicine, Luzerner Kantonsspital, University Teaching and Research Hospital, University of Lucerne, Switzerland (G.M.K.). UOC Neurologia e Stroke Unit, AORN Antonio Cardarelli, Naples, Italy (P. Candelaresi, V.A., C.M.). SCU Neurologia – Stroke Unit, Azienda Ospedaliera Universitaria Maggiore della Carità, Novara, Italy (R. Tarletti, A. Mele). Neurology Division (A. Ciacciarelli, M. Alessiani) and UOS UTN – Emergency Department (G. Monteforte), Santa Maria Goretti Hospital, Latina, Italy. Centre for Health Research, Department of Imaging Methods, Faculty of Medicine, University of Ostrava, Czech Republic (D.P.). Comprehensive Stroke Center, Department of Neurology, University Hospital Ostrava, Czech Republic (M. Roubec, D. Školoudík). Centre for Health Research, Department of Clinical Neurosciences, Faculty of Medicine, University of Ostrava, Czech Republic (M. Roubec, D. Školoudík). Department of Neurology (F. Vandervorst, S.D.R.) and Department of Radiology (M. Verdam), Universitair Ziekenhuis Brussel, Belgium. NEUR Research Group, Center for Neurosciences (C4N), Vrije Universiteit Brussel (VUB), Belgium (F. Vandervorst, S.D.R.). Stroke Unit, Department of Neurology, Military Faculty Hospital, Prague, Czech Republic (J.B., M. Šrámek). SC Neurologia – ASST Papa Giovanni XXIII, Bergamo, Italy (V.M., D.A.). SC Radiologia Diagnostica per Immagini 2 – Neuroradiologia ASST Papa Giovanni XXIII, Bergamo, Italy (L.Q.). SC Neurologia Stroke Unit, Dipartimento Neurologico e della Riabilitazione, ARNAS G. Brotzu, Cagliari, Italy (J. Moller). Stroke Center, Department of Systems Medicine, University of Rome Tor Vergata, Italy (I.M., M. Diomed, G.L.). Mercy St. Vincent Medical Center, Toledo, OH (O.O.Z., E.L., M.A.M.). Stroke Unit and Neurosonology Laboratory, Department of Neuroscience, Padua University Hospital, Italy (C.B., M. Zaccagnino, F. Viaro). Department of Medicine and Surgery (A.P.) and Department of Medicine and Surgery, Neurology Clinic (G. Avola, C.F.), University of Parma, Italy. Stroke Care Program Department of Emergency Parma University Hospital, Italy (A.P.). Department of Neurology and Clinical Neurophysiology, Collegium Medicum in Bydgoszcz, Nicolaus Copernicus University in Toruń, Poland (M. Świńska). Clinic of Neurology and Clinical Neurophysiology, Jan Bielecki University Hospital, Bydgoszcz, Poland (P.S.-S.). Department of Interventional Radiology, Jan Bielecki University Hospital No. 2, Bydgoszcz, Poland (G. Meder). Department of Neurology (L.M.-S., M. Knoflach) and Department of Radiology (E.R.G.), Medical University of Innsbruck, Austria. Beaumont Hospital, Royal College of Surgeons in Ireland, Dublin (M.R.C., P.N., S. Power). Clinical Department of Neurology, University Centre of Neurology and Neurosurgery, Faculty of Medicine, Wrocław Medical University, Poland (M.M.N.-K.). Department of General Radiology, Interventional Radiology, and Neuroradiology, Wrocław Medical University, Poland (M. Pula,

M.G.). Stroke Unit, Department of Neurology, Hospital Universitario Virgen Macarena, Sevilla, Spain (S.P.S., J. Montaner, A.S.B.). IRCCS Humanitas Research Hospital, Milan, Italy (S. Marcheselli, F. Vodret, C. Rappillo). Hospital Garcia de Orta, Unidade Local de Saúde de Almada-Seixal, Almada, Portugal (L. Pereira, M. Rodrigues). Department of Neurology, Na Homolce Hospital, Prague, Czech Republic (A.J., M. Kovar, J.V.). Department of Neurology, ASST Ovest Milanese, Legnano, Italy (G.V., S.G.C., A. Cardillo). Stroke Unit, UOC Neurologia Puca, AOU Consorziale Policlinico Bari, Italy (M. Petruzzellis, S.G., N.M.). Department of Neurology, University of Texas Rio Grande Valley – Valley Baptist, Harlingen (A.E.H., S. Miller). Department of Neurology, Washington University in St. Louis, MO (M.W.K.). Neurology Unit, Department of Neuro-Cardiovascular Medicine (V.P., S. Tonello) and Neuroradiology, Department of Neuro-Cardiovascular Medicine (S. Tollot), Treviso, Italy. Military Institute of Medicine – National Research Institute, Warsaw, Poland (A.D., J.S., Adam Stepień). Neurology Department and Stroke Unit, Hospital de Faro, ULS Algarve, Portugal (F.B., J. Ferrao). Neurology Unit, Stroke Unit, Azienda Unità Sanitaria Locale-IRCCS di Reggio Emilia, Italy (M. Zedde). Neuroradiology Unit, Ospedale Santa Maria della Misericordia, Rovigo, Italy (R.P.). SSD Stroke Unit, AOU Sassari, Italy (Alessandra Sanna, R.M.). Neurology Unit, Department of Medicine, Surgery and Pharmacy, University Hospital of Sassari, Italy (R.M.). Department of Medical Sciences and Public Health, University of Cagliari, Italy (R.M.). Department of Neurology and Interventional Neurology, University of Health Sciences, Antalya Training and Research Hospital, Turkey (K.D.). Department of Biotechnological and Applied Clinical Sciences, University of L'Aquila, Italy (S.S., M.G.V., R.O.). Department of Neurology and Stroke Unit (M. Mannino, V.T.) and Department of Radiology (M.F.), AOOR Villa Sofia-V. Cervello, Palermo, Italy. Department of Neuroradiology, S. Bortolo Hospital, Vicenza, Italy (M.F.). Neurosciences, University Hospitals of North Midlands NHS Trust, Stoke-on-Trent, Staffordshire, United Kingdom (E.H., C. Roffe). Keele University School of Medicine, Staffordshire, United Kingdom (E.H., C. Roffe). Department of Neurology, National Institute of Medicine Of the Ministry of Interior and Administration, Warsaw, Poland (M. Dorobek, J.Z.-T.). Neurology Unit, Luigi Sacco University Hospital, Milan, Italy (G. Scopelliti, L. Pantoni). Neuroscience Research Center, Department of Biomedical and Clinical Sciences, University of Milan, Milan, Italy (L. Pantoni). Department of Neurorehabilitation Sciences, Casa di Cura Igea, Milan, Italy (M.F.D., A.R.A.-B., R.G.N.). Department of Neurology, İstanbul University Faculty of Medicine, Turkey (M. Sezgin, Y.D., E.E.). Second Department of Neurology, National & Kapodistrian University of Athens, Attikon University Hospital, Greece (G.T., A. Theodorou). Neurology Department, Elias University Emergency Hospital, Bucharest, Romania (V.E.T., C.A.P., S. Petrescu). Neurology Department, Carol Davila University of Medicine and Pharmacy, Bucharest, Romania (V.E.T., C.A.P., S. Petrescu).

Sources of Funding

None.

Disclosures

Dr Al-Bayati reports compensation from Balt USA, LLC for consultant services; compensation from Scientia Vascular for consultant services; compensation from Imperative Care, Inc for consultant services; compensation from Stryker for consultant services; and compensation from Johnson & Johnson for consultant services. Dr Nogueira reports compensation from RapidPulse for consultant services; compensation from Biogen, Inc for consultant services; compensation from Corindus, Inc and Philips; stock options in Brainomix; compensation from Synchro for data and safety monitoring services; stock options in Unity Medical; compensation from Phenox for consultant services; compensation from Biogen, Inc for consultant services; compensation from Vesalio for consultant services; compensation from Cerenovus for consultant services; stock options in Corindus, Inc; compensation from Ceretrieve for consultant services; stock holdings in Quantanosis AI; stock options in Ceretrieve; compensation from Anaconda for consultant services; compensation from Takeda Pharmaceuticals for consultant services; stock options in Viseon, Inc; compensation from NeuroVasc Technologies, Inc for consultant services; compensation from Imperative Care for consultant services; compensation from Hybernia for consultant services; stock holdings in Brain4Care; compensation from Imperative Care, Inc for consultant services; compensation from Astrocyte for consultant services; compensation from Genentech for consultant services; compensation from Brainomix for consultant services; compensation from Prolong Pharmaceuticals for consultant services; compensation from Corindus Vascular Robotics for consultant services; compensation from Genentech for consultant services; compensation from Boehringer Ingelheim for consultant services; stock options in Perfuze; stock options in viz-AI; compensation from Cerenovus for consultant services; stock options in RapidPulse; compensation from Stryker Corporation for consultant services; stock holdings in Piraeus Medical; stock options in Vesalio; compensation from Perfuze for consultant services; stock options in viz-AI; compensation from Sensome for consultant services; stock options in Perfuze; compensation from Shanghai Wallaby for con-

sultant services; stock options in Cerebrotech; compensation from Brainomix for consultant services; compensation from Phenox, Inc for consultant services; compensation from viz-AI for consultant services; compensation from Vesalio for consultant services; compensation from Medtronic USA, Inc for consultant services; compensation from Ceretrieve for consultant services; grants from Stryker; compensation from Anaconda Biomed for consultant services; compensation from Medtronic USA, Inc for consultant services; compensation from Stryker for consultant services; stock options in Truvic; grants from Cerenovus; compensation from RapidPulse for consultant services; compensation from Merck KGaA for consultant services; compensation from Perfuze for consultant services; compensation from Prolong Pharmaceuticals for consultant services; compensation from viz-AI for consultant services; compensation from Cerebrotech for consultant services; and stock options in Reist/O'Apel Medical. Dr Zini reports compensation from Daiichi Sankyo for consultant services; compensation from Agenzia Italiana del Farmaco for other services; compensation from the Italian Health Ministry for other services; grants from the Italian Health Ministry; compensation from Bayer Healthcare for other services; compensation from Pfizer for consultant services; compensation from Boehringer Ingelheim for consultant services; compensation from CSL Behring for consultant services; compensation from Alexion Pharmaceuticals for consultant services; and compensation from Amgen for consultant services. Dr Scarcia reports compensation from Medtronic, Inc for expert witness services and compensation from Penumbra, Inc for expert witness services. Dr Mujanovic reports grants from Schweizerische Herzstiftung (Swiss Heart Foundation) and employment by University Hospital Bern, Inselspital. Dr Arnold reports compensation from Boehringer Ingelheim for consultant services; compensation from Covidien, Medtronic for consultant services; compensation from AstraZeneca, Bayer, Bristol Myers Squibb, Covidien, Daiichi Sankyo, Novartis, Sanofi, and Pfizer for consultant services; grants from the Swiss National Science Foundation; compensation from Amgen for consultant services; and compensation from Novo Nordisk for consultant services. Dr Al Kasab reports compensation from Stryker for other services; employment by the Medical University of South Carolina; and compensation from Medtronic for consultant services. Dr Sheth reports compensation from Phenox, Inc for data and safety monitoring services; compensation from Medtronic for consultant services; grants from the National Institutes of Health; compensation from Motif Neurosciences for other services; compensation from University of Iowa Hospitals & Clinics for other services; employment by UTHHealth McGovern Medical School; compensation from viz-AI for consultant services; compensation from Route 92 Medical, Inc for consultant services; compensation from Imperative Care, Inc for consultant services; and compensation from Penumbra, Inc for consultant services. Dr Wouters reports compensation from Bayer Healthcare for consultant services. Dr Tassi reports compensation from AstraZeneca for consultant services; compensation from Boehringer Ingelheim Pty, Ltd for consultant services; and compensation from Daiichi Sankyo for consultant services. Dr Min reports compensation from Medtronic for consultant services; grants from Medtronic USA, Inc; and grants from the Spectrum Health Foundation. Dr Świtońska reports compensation from Angelini Pharma for other services; compensation from AstraZeneca for other services; compensation from EVER Neuro Pharma for other services; compensation from AbbVie for consultant services; travel support from Merz Pharmaceuticals GmbH; compensation from Biogen for other services; and compensation from Teva Pharmaceutical Industries for other services. Dr Gizewski reports compensation from Medtronic for other services. Dr Nicholson reports compensation from CERENOVUS for consultant services; compensation from Stryker for consultant services; and compensation from Medtronic USA, Inc for consultant services. Dr Power reports compensation from Stryker for other services and compensation from Neuravi, Ltd for consultant services. Dr Kovar reports compensation from Bayer for other services. Dr Hassan reports employment by Valley Baptist Medical Center; compensation from Cerenovus for consultant services; compensation from MicroVention Inc for consultant services; compensation from Stryker Corporation for consultant services; compensation from GE Healthcare for consultant services; compensation from Medtronic for consultant services; compensation from viz-AI for consultant services; an ownership stake in Quantanosis.ai; and compensation from Penumbra, Inc for consultant services. Dr Sacco reports compensation from Boehringer Ingelheim for consultant services; compensation from Pfizer Canada, Inc for consultant services; compensation from H. Lundbeck A/S for consultant services; employment by Università degli Studi dell'Aquila; compensation from Novartis for consultant services; compensation from Teva Pharmaceutical Industries for consultant services; compensation from AstraZeneca for consultant services; compensation from Allergan for consultant services; compensation from Eli Lilly and Company for consultant services; and compensation from Novartis for other services. Dr Ornello reports compensation from Novartis for other services; travel support from Teva Pharmaceutical Industries; compensation from Eli Lilly and Company for other services; compensation from Teva Pharmaceutical Industries for consultant

services; compensation from Teva Pharmaceutical Industries for other services; grants from Pfizer; compensation from Organon for other services; compensation from H. Lundbeck A/S for other services; compensation from Bayer for other services; compensation from AbbVie for data and safety monitoring services; grants from Novartis; compensation from Eli Lilly for data and safety monitoring services; and grants from Allergan. Dr Roffe reports grants from Phenox, Inc. Dr Pantoni reports compensation from Medtronic for consultant services; compensation from PIAM for consultant services; compensation from Amicus Therapeutics for consultant services; compensation from Boehringer Ingelheim for consultant services; and compensation from Boston Scientific Corporation for consultant services. Dr Vlad Eugen Tiu reports compensation from Boehringer Ingelheim for consultant services. Dr Panea has nothing to disclose. Dr Nguyen reports compensation from the American Stroke Association for other services; compensation from Route 92 Medical, Inc for consultant services; compensation from Bayer for consultant services; and compensation from Medtronic for consultant services. Dr Sessa reports compensation from Boston Scientific Corporation for other services and compensation from Amgen for other services. The other authors report no conflicts.

Supplemental Material

Tables S1–S4

Figure S1

STROBE Checklist

REFERENCES

- Goyal M, Menon BK, van Zwam WH, Dippel DWJ, Mitchell PJ, Demchuk AM, Dávalos A, Majoie CBLM, van der Lugt A, de Miquel MA, et al; HERMES Collaborators. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. *Lancet*. 2016;387:1723–1731. doi: 10.1016/S0140-6736(16)00163-X
- Saver JL, Goyal M, van der Lugt A, Menon BK, Majoie CBLM, Dippel DWJ, Campbell BC, Nogueira RG, Demchuk AM, Tomasello A, et al; HERMES Collaborators. Time to treatment with endovascular thrombectomy and outcomes from ischemic stroke: a meta-analysis. *JAMA*. 2016;316:1279–1288. doi: 10.1001/jama.2016.13647
- Nogueira RG, Jadhav AP, Haussen DC, Bonafe A, Budzik RF, Bhuva P, Yavagal DR, Ribo M, Cognard C, Hanel RA, et al; DAWN Trial Investigators. Thrombectomy 6 to 24 hours after stroke with a mismatch between deficit and infarct. *N Engl J Med*. 2018;378:11–21. doi: 10.1056/NEJMoa1706442
- Lu R, Kiernan MC, Murray A, Rosner MH, Ronco C. Kidney-brain crosstalk in the acute and chronic setting. *Nat Rev Nephrol*. 2015;11:707–719. doi: 10.1038/nrneph.2015.131
- Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO Clinical Practice Guideline for acute kidney injury. *Kidney Int Suppl*. 2012;2:1–138.
- Schwarz G, Cascio Rizzo A, Ambler G, Wrona P, Slowik A, Kotas S, Doheim MF, Al-Bayati AR, Nogueira RG, Nunes AP, et al; as the CAN-REST Study Group. Contrast-associated acute kidney injury after thrombectomy for ischemic stroke: prognostic impact and CAN-REST predictive score. *Neurology*. 2026;106:e214655. doi: 10.1212/WNL.00000000000214655
- Oliveira M, Rocha AR, Barbosa F, Barros P, Fonseca L, Ribeiro M, Afreixo V, Gregório T. Acute kidney injury after endovascular therapy in acute stroke patients: systematic review with meta-analysis. *J Neurointerv Surg*. 2023;15:e468–e474. doi: 10.1136/jnis-2022-019955
- Diprose WK, Sutherland LJ, Wang MTM, Barber PA. Contrast-associated acute kidney injury in endovascular thrombectomy patients with and without baseline renal impairment. *J Neuroimaging*. 2020;30:316–320.
- Altman DG, Royston P. The cost of dichotomising continuous variables. *BMJ*. 2006;332:1080. doi: 10.1136/bmj.332.7549.1080
- Inker LA, Eneanya ND, Coresh J, Tighiouart H, Wang D, Sang Y, Crews DC, Doria A, Estrella MM, Froissart M, et al; Chronic Kidney Disease Epidemiology Collaboration. New creatinine- and cystatin C-based equations to estimate GFR without race. *N Engl J Med*. 2021;385:1737–1749. doi: 10.1056/NEJMoa2102953
- Sterne JAC, White IR, Carlin JB, Spratt M, Royston P, Kenward MG, Wood AM, Carpenter JR. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. *BMJ*. 2009;338:b2393. doi: 10.1136/bmj.b2393
- Cameron AC, Miller DL. A practitioner's guide to cluster-robust inference. *J Hum Resour*. 2015;50:317–372.
- Durrleman S, Simon R. Flexible regression models with cubic splines. *Stat Med*. 1989;8:551–561. doi: 10.1002/sim.4780080504
- Chertow GM, Burdick E, Honour M, Bonventre JV, Bates DW. Acute kidney injury, mortality, length of stay, and costs in hospitalized patients. *J Am Soc Nephrol*. 2005;16:3365–3370. doi: 10.1681/ASN.2004090740
- Lassnigg A, Schmidlin D, Mouhieddine M, Bachmann LM, Druml W, Bauer P, Hiesmayr M. Minimal changes of serum creatinine predict prognosis in patients after cardiothoracic surgery: a prospective cohort study. *J Am Soc Nephrol*. 2004;15:1597–1605. doi: 10.1097/01.asn.0000130340.93930.dd
- Kusirisin P, Chattipakorn SC, Chattipakorn N. Contrast-induced nephropathy and oxidative stress: mechanistic insights for better interventional approaches. *J Transl Med*. 2020;18:400. doi: 10.1186/s12967-020-02574-8
- Zhao Q, Yan T, Chopp M, Venkat P, Chen J. Brain-kidney interaction: renal dysfunction following ischemic stroke. *J Cereb Blood Flow Metab*. 2020;40:246–262. doi: 10.1177/0271678X19890931
- Poston JT, Koyner JL. Sepsis associated acute kidney injury. *BMJ*. 2019;364:k4891. doi: 10.1136/bmj.k4891
- Waikar SS, Betensky RA, Bonventre JV. Creatinine as the gold standard for kidney injury biomarker studies? *Nephrol Dial Transplant*. 2009;24:3263–3265. doi: 10.1093/ndt/gfp428
- Ronco C, Haapio M, House AA, Anavekar N, Bellomo R. Cardiorenal syndrome. *J Am Coll Cardiol*. 2008;52:1527–1539. doi: 10.1016/j.jacc.2008.07.051
- Delanaye P, Schaeffner E, Ebert N, Cavalier E, Mariat C, Krzesinski JM, Moranne O. Normal reference values for glomerular filtration rate: what do we really know? *Nephrol Dial Transplant*. 2012;27:2664–2672. doi: 10.1093/ndt/gfs265
- Royston P, Altman DG, Sauerbrei W. Dichotomizing continuous predictors in multiple regression: a bad idea. *Stat Med*. 2006;25:127–141. doi: 10.1002/sim.2331
- Sharp AJ, Patel N, Reeves BC, Angelini GD, Fiorentino F. Pharmacological interventions for the prevention of contrast-induced acute kidney injury in high-risk adult patients undergoing coronary angiography: a systematic review and meta-analysis of randomised controlled trials. *Open Heart*. 2019;6:e000864. doi: 10.1136/openhrt-2018-000864